



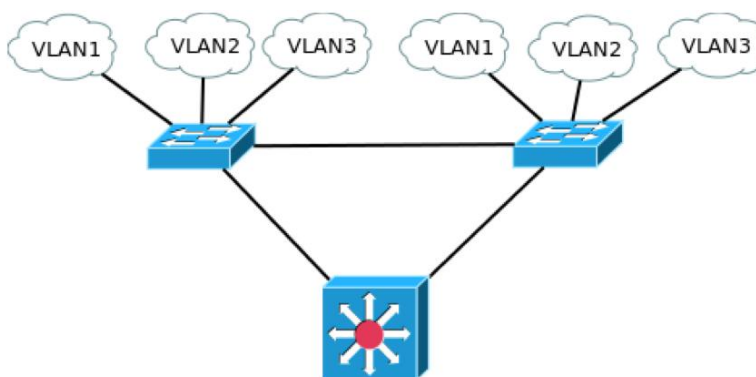
Guia 5– Interligação de VLANs

Objectivos:

- Routing entre VLANs
- Estabelecimento e configuração de trunks

1 - Inter-VLAN Routing: usando um Switch L3 (e links redundantes)

5. Troque o router por um switch L3 conforme a figura abaixo. Tenha em atenção quais as interfaces que usa para ligar o switch L3 aos outros!



Configure 3 VLANs no L3 Switch (VLAN1 ,2 e 3):

```
RouterSW# vlan database
RouterSW(vlan)# vlan 1
RouterSW(vlan)# vlan 2
RouterSW(vlan)# vlan 3
RouterSW(vlan)# exit
```

Configure os portos L2 do Switch L3 (fastEthernet slot 1/*), porto 0: VLAN1, portos 1-8: VLAN2, portos 9-12:

VLAN3 e portos 13-15: Inter-switch/Tagged/802.1Q:

```
RouterSW(config)# interface f1/0
RouterSW(config-if)# switchport mode access
RouterSW(config-if)# switchport access vlan 1
RouterSW(config-if)# interface range fastEthernet 1/1 - 8
RouterSW(config-if-range)# switchport mode access
RouterSW(config-if-range)# switchport access vlan 2
RouterSW(config-if-range)# interface range fastEthernet 1/9 - 12
RouterSW(config-if-range)# switchport mode access
```

Cofinanciado por:





```
RouterSW(config-if-range)# switchport access vlan 3
RouterSW(config-if-range)# interface range fastEthernet 1/13 - 15
RouterSW(config-if-range)# switchport mode trunk
RouterSW(config-if-range)# switchport trunk encapsulation dot1q

Configure as interfaces virtuais (Vlan) do switch L3:
RouterSW(config)# interface Vlan 1
RouterSW(config-if)# ip address 10.1.1.1 255.255.255.0
RouterSW(config-if)# no autostate !força o porto a estar sempre em cima
```

```
RouterSW(config)# interface Vlan 2
RouterSW(config-if)# ip address 10.2.2.1 255.255.255.0
RouterSW(config-if)# no autostate ! força o porto a estar sempre em cima
RouterSW(config)# interface Vlan 3
RouterSW(config-if)# ip address 10.3.3.1 255.255.255.0
RouterSW(config-if)# no autostate !forces the port to be always up
```

Verifique a tabela de encaminhamento (**show ip route**). Certifique-se que os terminais (hosts) têm as gateways configuradas para os endereços IP das interfaces virtuais (Vlan) do switch L3 e teste a conectividade. Capture os pacotes trocados entre o Router e o Switch L3.

6. Grave as configurações dos equipamentos, bem como o projecto.

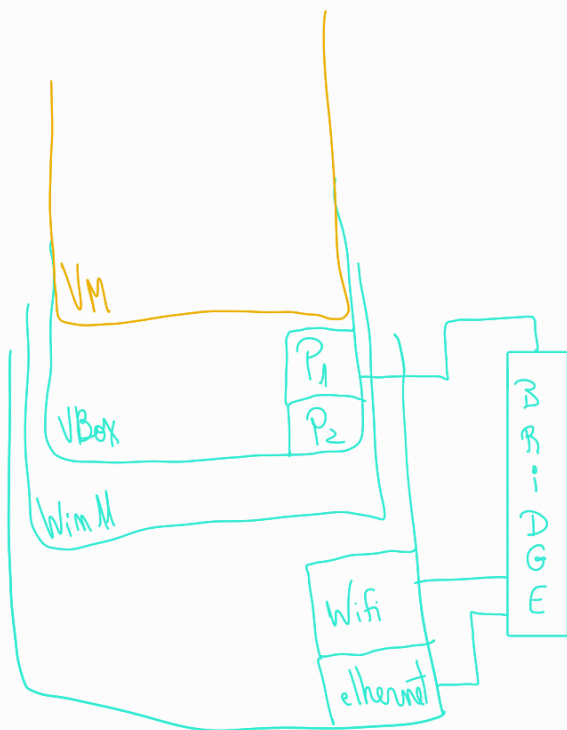
1 – Links de Trunk Restringidos

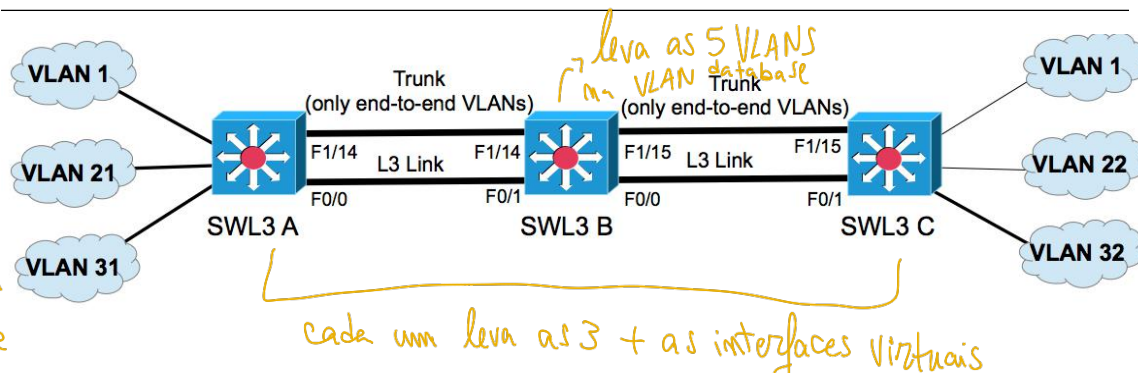
7. Construa a seguinte rede usando Switches L3. A VLAN 1 existe de uma ponta à outra da rede (end-to-end), e as outras VLANs são apenas locais. Para simular interfaces L3, use interfaces Ethernet normais, ou interfaces L2 (modulo de Switching) convertidas em interfaces L3 usando o commando no switchport.

Configure as VLANs nos três switches L3: SWL3 A deverá ter as VLAN 1, 21, and 31; SWL3 B deverá ter a VLAN 1 apenas; e o SWL3 C deverá ter as VLAN 1, 22, and 32.

Nota: Por defeito, os Cisco têm VLANs que têm que ser consideradas end-to-end (1002-1005) e não podem ser apagadas.

Nota: Não se esqueça que pode substituir as núvens por VPC's ou outros routers a fazer de host.





Configure as VLANs assumindo que os portos de trunk são f1/14 e f1/15 em todos os switches L3:

```
SWL3*(config-if-range)# interface range fastEthernet 1/14 - 15
```

```
SWL3*(config-if-range)# switchport mode trunk
```

```
SWL3*(config-if-range)# switchport trunk allowed vlan 1,1002-1005
```

Garanta que todos os switches L3 têm o routing IPv4 ativo, usando o comando `ip routing`. Configure as os endereços IP das interfaces assumindo que as VLANs têm as redes 10.0.<#vlan>.0/24, a rede IP entre o SWL3A e o SWL3B é a 10.1.0.0/24 e a rede entre o SWL3B e o SWL3C é a 10.2.0.0/24.

Configure um protocolo de encaminhamento.

```
SWL3*(config)# router rip
```

```
SWL3*(config-router)# version 2
```

!Necessário devido às sub-redes

```
SWL3*(config-router)# network 10.0.0.0
```

```
SWL3*(config-router)# passive-interface vlan 1
```

! Não utiliza a VLAN1 para encaminhar trafego L3

Analise as tabelas de encaminhamento, teste a conectividade e capture/analise as capturas no trunk e nos links L3.

Nota:

Para fazer um ping desde a interface da VLAN1 no SWL3A (e.g., 10.0.1.1) para a interface VLAN22 do SWL3C (e.g., 10.0.22.3), utilize o commando ping com indicação da origem:

```
SWL3A# ping 10.0.22.3 source 10.0.1.1
```

Sem o comando source o endereço origem do ping é o da interface de saída definida pela tabela de encaminhamento.

Conclua acerca de que tráfego passa através de que link, entre os switches, quando existe conectividade inter-VLAN.

8. Grave as configurações dos equipamentos, bem como o projecto.

2 – Links de Trunk Restringidos (sem Link L3)

9. Remova o link L3, crie uma VLAN de interligação (e.g., VLAN 101) e reconfigure a rede para ter conectividade total entre VLANs. Comece por alterar as restrições do trunk:

```
SWL3*(config-if-range)# switchport trunk allowed vlan 1,101,1002-1005
```

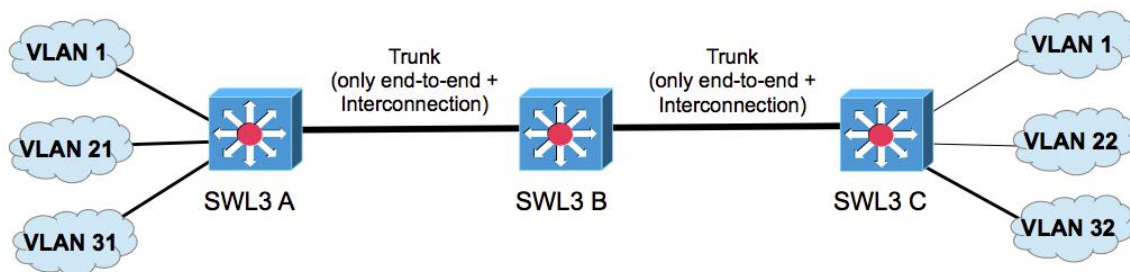
Cofinanciado por:





Analise as tabelas de encaminhamento, teste a conectividade e capture/analise as capturas nos trunks.

DICA: As VLANs de interligação podem ser usadas para passar anúncios de rotas RIP de outras redes.



10. Grave as configurações dos equipamentos, bem como o projecto.

ssh user_destino@IP

scp origem destino

scp fich.zip user@IP:/home/user

scp user@IP:/home/user .

scp -R pg@192.168.229.238:/home/pg/Docencia

route -n (para ver a routing table)

https://labserver.estga.ua.pt/CiscoGNS3.zip

https://192.168.229.2/CiscoGNS3.zip

ligar à rede AP2026 (emAgueda,26)

https://192.168.229.2/IOUs.zip

cd /home/\$(USER)/GNS3/images/IOU

chmod +x *.bin

```
ines@ines:~/Documents/PR_PlaneamentoDeRedes/IOU$ history | grep scp
 79  scp pg@192.168.229.238:/home/pg/C* .
 81  scp pg@192.168.229.238:/home/pg/C* .
150  history | grep scp
ines@ines:~/Documents/PR_PlaneamentoDeRedes/IOU$ !79
scp pg@192.168.229.238:/home/pg/C* .
The authenticity of host '192.168.229.238 (192.168.229.238)' can't be established.
ED25519 key fingerprint is SHA256:IT1D2G4RdMnboZAZpsdeEexsZRXaPV1KAXoRng5cmgA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.229.238' (ED25519) to the list of known hosts.
pg@192.168.229.238's password:
scp: download /home/pg/CheckPoint/: not a regular file
ines@ines:~/Documents/PR_PlaneamentoDeRedes/IOU$ scp pg@192.168.229.238:/home/pg/C* .
pg@192.168.229.238's password:
scp: download /home/pg/CheckPoint/: not a regular file
CiscoKeyGen.py                                100% 1621   101.6KB/s   00:00
ines@ines:~/Documents/PR_PlaneamentoDeRedes/IOU$ python3 CiscoKeyGen.py

*****
Cisco IOU License Generator - Kal 2011, python port of 2006 C version
hostid=007f0101, hostname=ines, ioukey=7f02b0

*****
Create the license file $HOME/.iourc with this command:
echo -e '[license]\nines = ae90ef5c0697d4d6;' | tee $HOME/.iourc

The command adds the following text to $HOME/.iourc:
[license]
ines = ae90ef5c0697d4d6;

*****
Disable the phone home feature with this command:
grep -q -F '127.0.0.1 xml.cisco.com' /etc/hosts || echo '127.0.0.1 xml.cisco.com' | sudo
tee -a /etc/hosts

The command adds the following text to /etc/hosts:
127.0.0.1 xml.cisco.com

*****
```

VM Proxmox, na ESTGA pelo navegador:

<https://192.168.229.1:8006>

user: ines

pass: 321654

Com a VPN da UA, em casa:

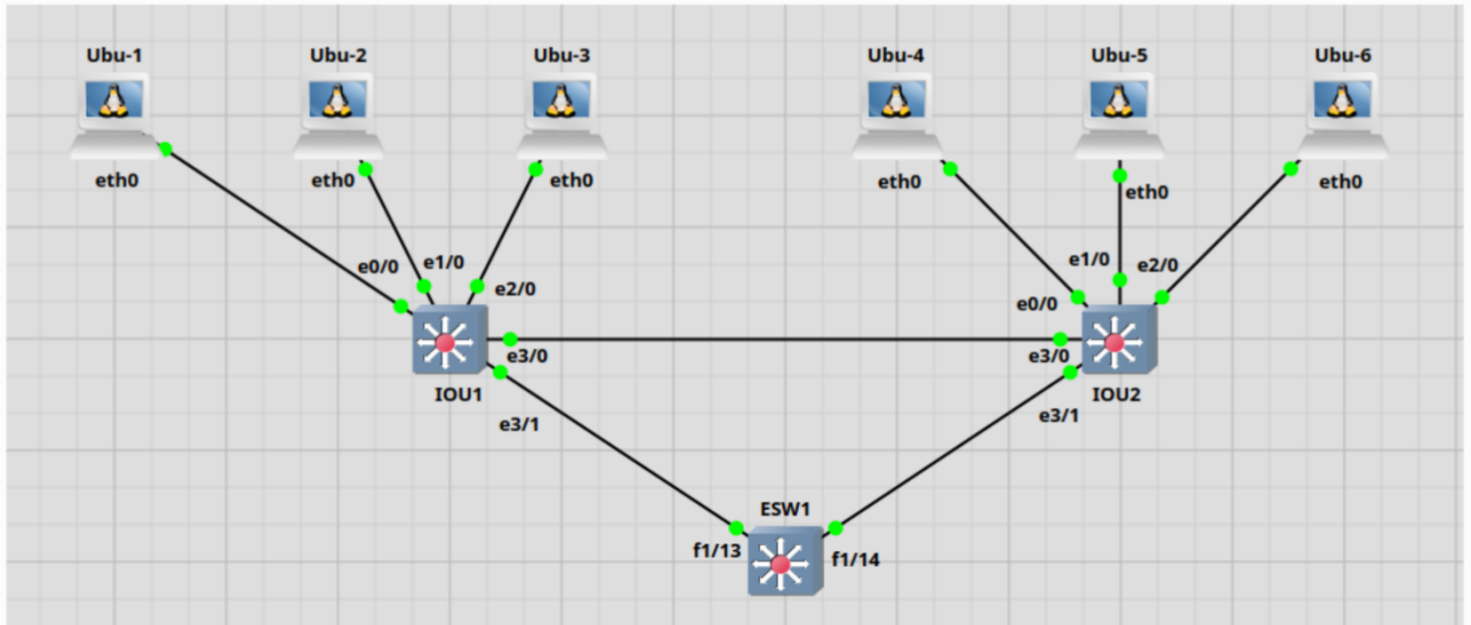
<https://192.168.100.249:1080>

realm: pve

Inter-VLAN Routing: usando um Switch L3 (e links redundantes)

5.

Topologia da rede:



Configuração do Switch de Layer 3 ESW1:

```
ESW1#sh run
Building configuration...
```

```
Current configuration : 3096 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
no service dhcp
!
hostname ESW1
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
memory-size iomem 5
no ip icmp rate-limit unreachable
ip cef
```



```
!  
!  
!  
!  
no ip domain lookup  
ip auth-proxy max-nodata-conns 3  
ip admission max-nodata-conns 3
```

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
vtp file nvram:vlan.dat
```

```
!  
!  
ip tcp synwait-time 5
```

```
!  
!  
!  
!  
!  
!  
interface FastEthernet0/0  
description *** Unused for Layer2 EtherSwitch ***  
no ip address  
shutdown  
duplex auto  
speed auto
```

```
!  
interface FastEthernet0/1  
description *** Unused for Layer2 EtherSwitch ***  
no ip address  
shutdown  
duplex auto  
speed auto
```

```
!  
interface FastEthernet1/0  
duplex full  
speed 100
```

```
!  
interface FastEthernet1/1
```

```
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/2
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/3
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/4
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/5
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/6
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/7
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/8
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/9
switchport access vlan 3
duplex full
speed 100
!
interface FastEthernet1/10
switchport access vlan 3
duplex full
speed 100
!
interface FastEthernet1/11
switchport access vlan 3
```

```
duplex full
speed 100
!
interface FastEthernet1/12
switchport access vlan 3
duplex full
speed 100
!
interface FastEthernet1/13
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet1/14
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet1/15
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet2/0
no ip address
shutdown
duplex auto
speed auto
!
interface Vlan1
ip address 10.1.1.1 255.255.255.0
no autostate
!
interface Vlan2
ip address 10.2.2.1 255.255.255.0
no autostate
!
interface Vlan3
ip address 10.3.3.1 255.255.255.0
no autostate
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
```

```
!  
control-plane  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
banner exec ^C
```

This is a normal Router with a SW module inside (NM-16ESW)
It has been preconfigured with hard coded speed and duplex

To create vlans use the command "vlan database" from exec mode
After creating all desired vlans use "exit" to apply the config

To view existing vlans use the command "show vlan-switch brief"

Warning: You are using an old IOS image for this router.
Please update the IOS to enable the "macro" command!

```
^C  
!  
line con 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous  
line aux 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous  
line vty 0 4  
  login  
!  
!  
end
```

Há conectividade entre todas as VLANs:

```
root@Ubu-1:~# ping 10.1.1.2
PING 10.1.1.2 (10.1.1.2) 56(84) bytes of data.
64 bytes from 10.1.1.2: icmp_seq=1 ttl=64 time=0.020 ms
64 bytes from 10.1.1.2: icmp_seq=2 ttl=64 time=0.055 ms
^C
--- 10.1.1.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 0.020/0.037/0.055/0.017 ms
root@Ubu-1:~# ping 10.1.1.3
PING 10.1.1.3 (10.1.1.3) 56(84) bytes of data.
64 bytes from 10.1.1.3: icmp_seq=1 ttl=64 time=7.57 ms
64 bytes from 10.1.1.3: icmp_seq=2 ttl=64 time=1.58 ms
64 bytes from 10.1.1.3: icmp_seq=3 ttl=64 time=2.18 ms
64 bytes from 10.1.1.3: icmp_seq=4 ttl=64 time=10.8 ms
64 bytes from 10.1.1.3: icmp_seq=5 ttl=64 time=1.65 ms
64 bytes from 10.1.1.3: icmp_seq=6 ttl=64 time=1.54 ms
^C
--- 10.1.1.3 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5010ms
rtt min/avg/max/mdev = 1.537/4.213/10.756/3.625 ms
root@Ubu-1:~# ping 10.2.2.3
PING 10.2.2.3 (10.2.2.3) 56(84) bytes of data.
64 bytes from 10.2.2.3: icmp_seq=2 ttl=63 time=16.9 ms
64 bytes from 10.2.2.3: icmp_seq=3 ttl=63 time=14.6 ms
64 bytes from 10.2.2.3: icmp_seq=4 ttl=63 time=19.0 ms
^C
--- 10.2.2.3 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3037ms
rtt min/avg/max/mdev = 14.601/16.823/19.008/1.799 ms
root@Ubu-1:~# ping 10.3.3.3
PING 10.3.3.3 (10.3.3.3) 56(84) bytes of data.
64 bytes from 10.3.3.3: icmp_seq=2 ttl=63 time=21.4 ms
64 bytes from 10.3.3.3: icmp_seq=3 ttl=63 time=22.3 ms
64 bytes from 10.3.3.3: icmp_seq=4 ttl=63 time=19.0 ms
^C
--- 10.3.3.3 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3043ms
rtt min/avg/max/mdev = 18.974/20.878/22.300/1.399 ms
```

```
ESW1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

 10.0.0.0/24 is subnetted, 3 subnets
C       10.3.3.0 is directly connected, Vlan3
C       10.2.2.0 is directly connected, Vlan2
C       10.1.1.0 is directly connected, Vlan1
```

Configuração do switch Layer 2, IOU1:

IOU1#sh run

Building configuration...

Current configuration : 2124 bytes

!

! Last configuration change at 21:01:54 WET Fri Feb 27 2026

!

version 15.2

service timestamps debug datetime msec

```
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname IOU1
!
boot-start-marker
boot-end-marker
!
!
logging discriminator EXCESS severity drops 6 msg-body drops EXCESSCOLL
logging buffered 50000
logging console discriminator EXCESS
!
no aaa new-model
clock timezone WET 0 0
!
!
!
!
!
no ip icmp rate-limit unreachable
!
!
!
no ip domain-lookup
ip cef
no ipv6 cef
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
ip tcp synwait-time 5
!
!
!
!
!
!
!
!
!
!
!
interface Ethernet0/0
```

```
switchport mode access
!
interface Ethernet0/1
switchport mode access
!
interface Ethernet0/2
switchport mode access
!
interface Ethernet0/3
switchport mode access
!
interface Ethernet1/0
switchport access vlan 2
switchport mode access
!
interface Ethernet1/1
switchport access vlan 2
switchport mode access
!
interface Ethernet1/2
switchport access vlan 2
switchport mode access
!
interface Ethernet1/3
switchport access vlan 2
switchport mode access
!
interface Ethernet2/0
switchport access vlan 3
switchport mode access
!
interface Ethernet2/1
switchport access vlan 3
switchport mode access
!
interface Ethernet2/2
switchport access vlan 3
switchport mode access
!
interface Ethernet2/3
switchport access vlan 3
switchport mode access
!
interface Ethernet3/0
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Ethernet3/1
switchport trunk encapsulation dot1q
switchport mode trunk
!
```

```

interface Ethernet3/2
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface Ethernet3/3
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface Vlan1
  no ip address
  shutdown
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
!
!
!
!
!
!
control-plane
!
!
line con 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  login
!
!
end

```

Configuração do switch Layer 2, IOU2:

```

IOU2#sh run
Building configuration...

Current configuration : 2124 bytes
!
! Last configuration change at 21:01:54 WET Fri Feb 27 2026
!
version 15.2
service timestamps debug datetime msec

```



```
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname IOU2
!
boot-start-marker
boot-end-marker
!
!
logging discriminator EXCESS severity drops 6 msg-body drops EXCESSCOLL
logging buffered 50000
logging console discriminator EXCESS
!
no aaa new-model
clock timezone WET 0 0
!
!
!
!
!
no ip icmp rate-limit unreachable
!
!
!
no ip domain-lookup
ip cef
no ipv6 cef
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
ip tcp synwait-time 5
!
!
!
!
!
!
!
!
!
!
!
interface Ethernet0/0
```

```
switchport mode access
!
interface Ethernet0/1
switchport mode access
!
interface Ethernet0/2
switchport mode access
!
interface Ethernet0/3
switchport mode access
!
interface Ethernet1/0
switchport access vlan 2
switchport mode access
!
interface Ethernet1/1
switchport access vlan 2
switchport mode access
!
interface Ethernet1/2
switchport access vlan 2
switchport mode access
!
interface Ethernet1/3
switchport access vlan 2
switchport mode access
!
interface Ethernet2/0
switchport access vlan 3
switchport mode access
!
interface Ethernet2/1
switchport access vlan 3
switchport mode access
!
interface Ethernet2/2
switchport access vlan 3
switchport mode access
!
interface Ethernet2/3
switchport access vlan 3
switchport mode access
!
interface Ethernet3/0
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Ethernet3/1
switchport trunk encapsulation dot1q
switchport mode trunk
!
```

```
interface Ethernet3/2
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface Ethernet3/3
  switchport trunk encapsulation dot1q
  switchport mode trunk
!
interface Vlan1
  no ip address
  shutdown
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
!
!
!
!
!
!
control-plane
!
!
line con 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line aux 0
  exec-timeout 0 0
  privilege level 15
  logging synchronous
line vty 0 4
  login
!
!
end
```

Links de Trunk Restringidos

7.

Links de Trunk Restringidos (sem Link L3)

9.

